Problem Set 5

AI 501: Mathematics for Artificial Intelligence | Fall 2026

Problem 1

Let

$$f(x) = \tfrac{1}{2} x^T A x + b^T x,$$

where $A \in \mathbb{R}^{n \times n}$ is symmetric and $b \in \mathbb{R}^n$.

(a) Find the gradient $\nabla f(x)$. *Hint:* differentiate with respect to a single coordinate x_k and use the symmetry of A.

(b) Write the least-squares loss $\mathcal{L}(\theta) = \frac{1}{2}\|y - X\theta\|_2^2$ in the form above (up to a constant), and use part (a) to find its gradient $\nabla \mathcal{L}(\theta)$. *Hint:* expand the squared norm as an inner product.

Problem 2

The partial derivatives of $\|Ax\|_2^2$ with respect to $x_1, \ldots, x_n$ form the vector $2A^T Ax$, and the partial derivatives of $2b^T Ax$ form the vector $2A^T b$.

(a) Use these facts to find the condition under which all partial derivatives of $\|Ax - b\|_2^2$ are zero.

(b) Assuming the columns of A are linearly independent, write the minimizer in closed form, and relate it to the linear regression solution $\theta = (X^T X)^{-1} X^T y$.

Problem 3

We wish to fit a model with only a bias term, $\hat{y} = \theta_0$, to the four observations $y = (0, 8, 8, 20)^T$. An exact fit would require

$$\theta_0 = 0, \qquad \theta_0 = 8, \qquad \theta_0 = 8, \qquad \theta_0 = 20,$$

which is impossible.

(a) Write this system as $X\theta = y$, identifying the 4×1 design matrix X.

(b) Solve the normal equation $X^T X\hat{\theta} = X^T y$ for θ_0.

(c) Find the residual vector $e = y - X\hat{\theta}$ and $\|e\|_2^2$.

(d) What does this tell you about the least-squares prediction of a model with no input features?

Problem 4

Fit the line $\hat{y} = \theta_0 + \theta_1 x$ by least squares to the data points

$$(x, y) = (-1, 7), \quad (1, 7), \quad (2, 21).$$

(a) Write the three equations $X\theta = y$, identifying X and y.

(b) Solve the normal equations for $\hat{\theta} = (\theta_0, \theta_1)^T$.

(c) Find the fitted values $\hat{y} = X\hat{\theta}$ and the residual vector $e = y - \hat{y}$.

(d) Verify that e is orthogonal to both columns of X.

Problem 5

Find the best least-squares line $\hat{y} = \theta_0 + \theta_1 x$ for the data

$$y = (4, 2, -1, 0, 0)^T \quad \text{at} \quad x = -2, -1, 0, 1, 2.$$

(a) Compute $X^T X$ and $X^T y$, and solve for $\hat{\theta}$.

(b) Find the fitted values, the residual vector e, and $\|e\|_2^2$.

(c) Explain why X^TX is diagonal for this data set, and what this means for solving for θ_0 and θ_1.

Problem 6

(a) Find the degree-2 polynomial $\hat{y} = \theta_0 + \theta_1 x + \theta_2 x^2$ that passes through the points $(x, y) = (0, 6),\ (1, 0),\ (2, 0)$.

(b) What are the fitted values and the residuals here? Why are the normal equations unnecessary?

(c) A fourth data point (x_4, y_4) is added that does not lie on this parabola. Will the new least-squares parabola keep zero residual at the first three points and put all the error at the fourth? Explain.

Problem 7

Fit the degree-2 polynomial $\hat{y} = \theta_0 + \theta_1 x + \theta_2 x^2$ by least squares to the data

$$y = (0, 0, 1, 0, 0)^T \quad \text{at} \quad x = -2, -1, 0, 1, 2.$$

(a) Write the five equations $X\theta = y$ for the polynomial to pass through all five points. Explain why there is no exact solution.

(b) Compute X^TX. Before solving, predict the value of θ_1 and justify your prediction. *Hint:* notice what the symmetry of the inputs does to the power sums in X^TX.

(c) Solve the normal equations $X^TX\hat{\theta} = X^Ty$.

(d) Find the residual vector $e = y - X\hat{\theta}$ and $\|e\|_2^2$. Verify that e is orthogonal to every column of X.

Problem 8

A data set of $n = 100$ observations has a single input feature X and a numerical output Y. Two models are fit by least squares: a linear model $Y = \beta_0 + \beta_1 X + \varepsilon$, and a cubic model $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \varepsilon$. For each model, let e_{train} be the vector of residuals $y_i - \hat{y}_i$ on the training data and e_{test} the vector of residuals on a separate test set.

(a) Suppose the true relationship between X and Y is linear. Compare the training error $\|e_{\text{train}}\|_2^2$ of the two models. Would you expect one to be lower, would you expect them to be equal, or is there not enough information to tell? Justify your answer.

(b) Answer (a) for the test error $\|e_{\text{test}}\|_2^2$ instead of $\|e_{\text{train}}\|_2^2$.

(c) Suppose the true relationship is not linear, but you do not know how far from linear it is. Compare $\|e_{\text{train}}\|_2^2$ for the two models, with justification.

(d) Answer (c) for $\|e_{\text{test}}\|_2^2$ instead of $\|e_{\text{train}}\|_2^2$.